MILITARY SPECIFICATION

INDICATOR, VERTICAL VELOCITY AAU-18/A, INTEGRALLY LIGHTED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a 2-inch red and white integrally lighted, vertical velocity indicator having a range of from 0 to 6,000 fpm climb and descent, designated AAU-18/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloy
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

FEDERAL

FED-STD-595	Colors

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MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-882	System Safety Program Requirements
MS26105	Cover Glass, Aircraft Instrument Dial
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33565	Pointers, Dial, Standard Design of Aircraft Instrument
MS33638	Cases, Instrument, Flanged Mounted, Aircraft
MS33649	Bosses, Fluid Connection - Internal Straight Thread

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to the applicable specifications specified herein. Materials which are not covered by applicable specifications shall be of the best commercial quality and suitable for the purpose intended.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

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3.3.2 Metals. Metal shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Requirement 16 of MIL-STD-454.

3.3.3 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. Materials, as installed in the indicator and under the conditions specified herein shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.3.4 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction

3.4.1 Design. The indicator shall be as shown on figure 1 and shall be of such design as to permit overhaul and repair without the use of special tools and fixtures unless they are approved by the procuring activity. Special tools and commercial standard tools are defined in MIL-S-8512.

3.4.2 Construction. The indicator shall be so constructed as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.3 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,200 hours demonstrated to a 90 percent confidence level.

3.4.4 Maintainability. Maintainability requirements shall be in accordance with MIL-STD-470 for Class 1D equipment.

3.4.5 Safety. The indicator shall present no danger, injury, or hazard to operating and service personnel.

3.4.5.1 Hazard analysis. A safety hazard analysis shall be performed to determine, from a safety consideration, the functional relationships of components and equipments interfacing with the indicator. The safety analysis shall identify all components, equipments, and materials whose performance degradation of functional failure could result in category II and I hazards as defined in MIL-STD-882A, paragraph entitled Hazard Level. This analysis shall include a determination of the failure modes and the effects on safety should failures occur. Areas to be considered are the appropriate sections of MIL-STD-882A, as listed under paragraph entitled System Hazard Analyses, and the effects of storage, shelf life, transportation, and packaging. Where

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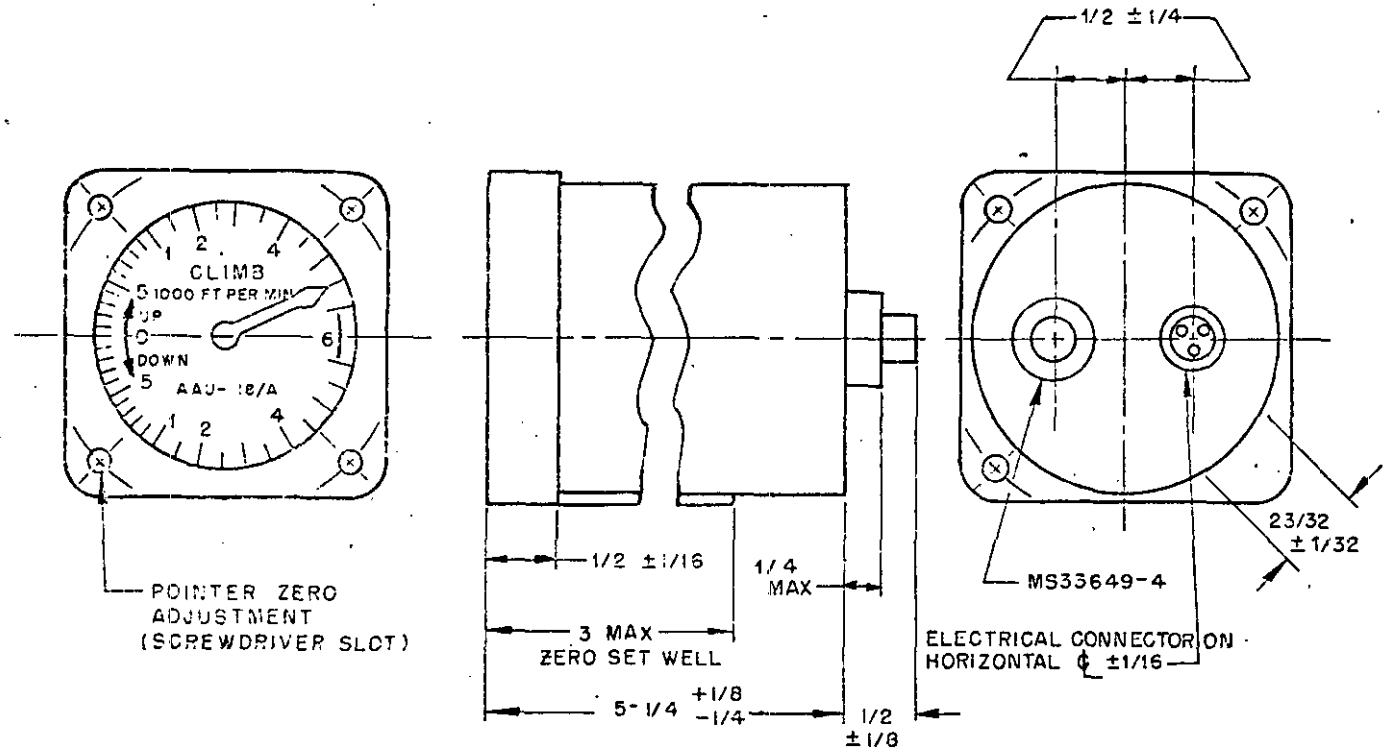


FIGURE 1. Case.

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practicable, design changes shall be made to eliminate or minimize the hazards. If the hazards cannot be eliminated, alternative controls such as recommended changes to interfacing equipment shall be presented to the procuring activity for resolution.

3.5 Performance. The indicator shall operate satisfactorily when subjected to the following conditions:

a. Temperature: Temperatures ranging from -65° to +71°C

b. Vibration

(1) Vibration error: Vibration at a double amplitude of 0.003 to 0.005 inch at a frequency of 5 to 50 Hz

(2) Vibration endurance: Vibration at a double amplitude of 0.018 to 0.02 inches at a frequency of 500 to 3000 cpm.

c. Humidity: Relative humidity up to 100 percent.

d. Salt fog: Exposure to simulated salt sea atmosphere for a period of 50 hours.

e. Fungus resistance: Exposure to fungus growth as encountered in tropical climates.

f. Overpressure: Overpressure within the range specified in 4.6.9.

g. Altitude: Altitudes ranging from sea level to 50,000 feet (3.436 inches Hg).

3.5.1 Magnetic effect. When the indicator is revolved about a short bar magnetic compass with a field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1°.

3.5.2 Leakage. Leakage shall not exceed 0.025 inch Hg per minute when the indicator is subjected to a vacuum approximately equivalent to 40,000 feet altitude.

3.5.3 Scale error. Scale error shall not exceed the limits as specified in table I.

3.5.4 Position error. Position error shall not exceed 50 fpm in any position.

3.5.5 Pointer lag. Pointer lag shall be not less than 4 seconds nor more than 7 seconds.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Case. The case shall consist of a body and bezel ring and shall be in accordance with MS33638, for the 2-inch nominal size, except as shown on figure 1. The case shall be airtight except for the static pressure connection.

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TABLE I. Tolerances.

Indicator Test Setting fpm	Altitude Interval ft	Tolerance Room Temp Ascent and Descent fpm	Tolerance Low (-35°C) and High (+45°C) Temp Ascent and Descent fpm
500	1,000 to 1,500	100	100
1,000	1,000 to 2,000	200	200
2,000		300	300
3,000	2,000 to 4,000	300	---
4,000		400	400
5,000		500	---
2,000	15,000 to 17,000	300	300
4,000		400	400
2,000	28,000 to 30,000	300	300
4,000		400	---

3.7.1 Body. The body shall be made of nonferrous, low-density metal, shall be uniform in texture, shall have a smooth surface, and shall be entirely covered with a durable black finish. The mounting-flange portion of the body shall have a lusterless finish.

3.7.2 Bezel ring. The bezel ring shall be made of aluminum alloy and shall have a durable, lusterless black finish. The bezel ring shall be held in place by means of screws properly secured with lockwashers.

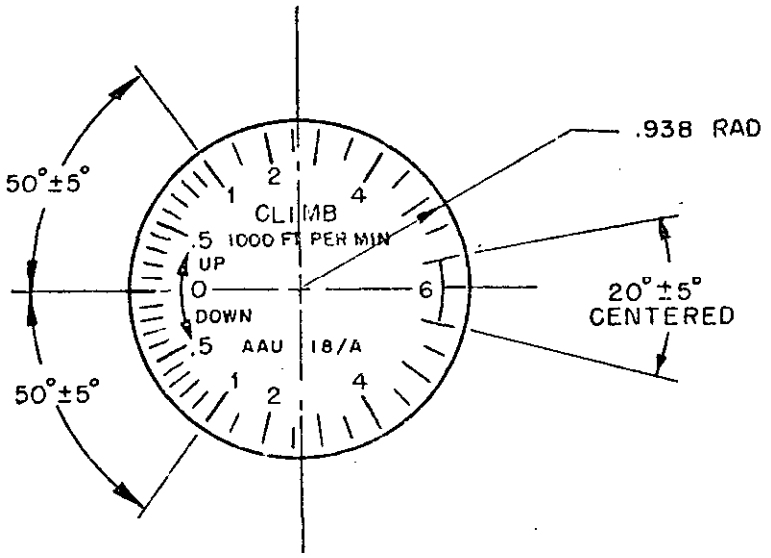
3.7.3 Cover glass. The cover glass shall be in accordance with MS28105-7 and shall be properly sealed. It shall be replaceable by removing the bezel ring. If the cover glass is used as part of the lighting system, its size and shape may deviate from the dimensions in accordance with MS28105-7 to the extent required for proper lighting design. If the cover glass and bezel ring form an integral assembly, they may be replaced as a unit.

3.7.3.1 Coverglass and lighting wedge. All reflecting glass surfaces shall be provided with a reflection-reducing coating and shall be in accordance with MIL-C-14806.

3.7.4 Dial. The dial size shall be 1.875 inch and shall be as shown on figure 2.

3.7.4.1 Letters and numerals. The dial letters and numerals shall be in accordance with MS33558.

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DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.
TOLERANCES: DECIMALS ±0.015, ANGLES ±0°15'

FIGURE 2. Dial.

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3.8 Operating mechanism. The operating mechanism shall be of the leak type, and shall be sufficiently rugged to prevent slight distortion of the case from affecting the accuracy of the instrument.

3.9 Zero setting system. A zero setting system controlled by a slotted setting shaft or screw, shall be provided that will permit the instrument to be set to indicate zero. Use of the zero setting system shall not change the calibration of the instrument.

3.10 Visibility of dial. The pointer, numerals, at least .0625 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone the side of which makes an angle of 20° with a perpendicular to the dial. The small diameter of the cone shall be the aperture of the instrument case.

3.11 Pointer. The pointer shall be in accordance with MS33585-3. It shall be light and sufficiently rigid to prevent oscillation under vibration and shall be firmly attached to the mechanism. Suitable stops shall be incorporated within the mechanism to limit pointer movement. The pointer shall rotate to the 6,000-foot graduation, ±.0312 inch, on both ascending and descending scales.

3.12 Electrical connector. The electrical connector shall be Deutsch connector DM5600-3P, or equal, and shall be located on the rear of the case as shown on figure 1.

3.13 Integral lighting. The indicator shall be integrally lighted with both red and white lighting circuits as shown on figure 3. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. Lighting shall operate on 5V dc or 5V ac. Lighting acceptability of production indicators shall be based on visual comparison with a prime standard. The method of lighting shall be approved by the procuring activity prior to fabrication.

3.13.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas covering the entire face of the indicator shall be measured for brightness level in foot-lamberts. A minimum of 10 areas covering the entire face of the indicator shall be measured for color. The prime standard instrument and its recorded overall brightness and color measurements shall be approved by the procuring activity. The standard shall be checked at least every 100 hours of operation to determine any change in brightness and color. Variations greater than ±20 percent in two or more areas shall require return of the standard for corrective action.

3.14 Synthetic jewel bearings. Synthetic jewel bearings shall be in accordance with MIL-B-27497.

3.15 Screw threads. The threads of all machine screws shall be in accordance with MIL-S-7742.

3.16 Weight. The weight of the indicator, completely assembled but without shipping plug and mounting screws, shall not exceed 1.75 pounds.

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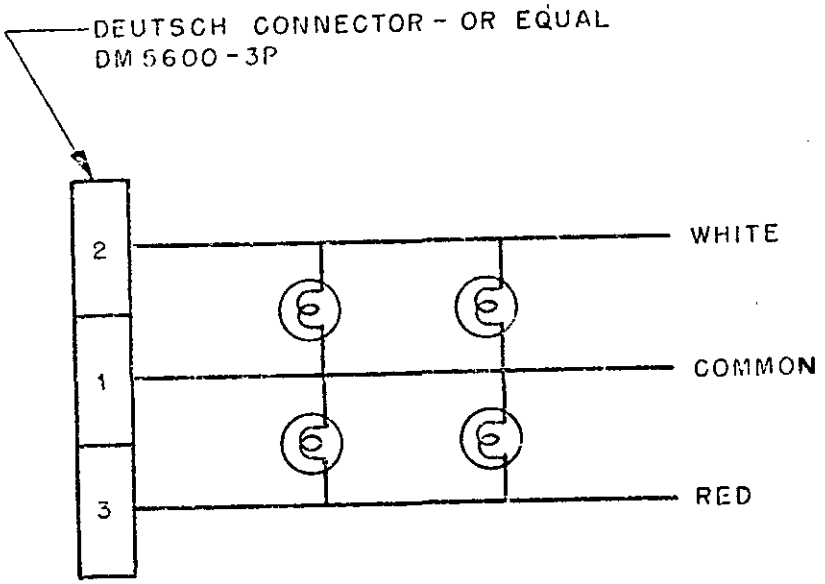


FIGURE 3. Electrical connector.

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3.17 Markings. The following markings shall be finished in lusterless white, Color No. 37875 of FED-STD-595. The dial and pointer marking dimensions shall be as follows:

	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005
Zero graduation	0.188	0.093
1,000 fpm graduations	0.188	0.030
100 fpm graduations	0.126	0.015
500 fpm graduations	0.126	0.030
Arc connecting 6,000 fpm marks	- - -	0.0312
Numeral .5	0.125	- - - -
Numerals 0, 1, 2, 4, 6	0.188	- - - -
Lettering CLIMB	0.125	- - - -
Shaded part of pointer	- - -	- - - -
Lettering UP and DOWN	0.094	- - - -
Lettering 1,000 FEET PER MINUTE	0.047	0.015

3.17.1 The following markings shall be finished in durable lusterless black, Color No. 37038 of FED-STD-595. The dimensions shall be as follows:

	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005
AAU-18/A	0.047	- - - -
Unshaded part of pointer and background of dial	- - -	- - - -

3.18 Mounting screws. The contractor shall furnish sufficient .138-32 UNC-2A round-head brass machine screws, having a durable lusterless black oxidized or black-nickel finish. The length shall be sufficient for mounting on a panel .1875 inch thick.

3.18.1 Envelope. When required, the contractor shall furnish a pack with each instrument an envelope containing mounting screws. The envelope shall be marked with the following information:

IMPORTANT

THIS ENVELOPE CONTAINS
INSTRUCTIONS, MOUNTING
SCREWS, AND SPRING-NUTS

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3.19 Finishes and protective coatings

3.19.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film in accordance with MIL-A-8625 except as follows:

3.19.1.1 Dials, small holes, and case inserts need not be anodized.

3.19.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.19.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.19.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.19.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.19.2 Steel parts. Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III as applicable, and of a class that is adequate to achieve the degree of protection required.

3.20 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.21 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.21.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification tests

b. Quality assurance tests.

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25oC). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the instrument shall be tested in the normal operating position and shall be lightly tapped or vibrated before a test reading is taken. The magnitude of vibration applied shall be sufficient to remove all friction from the indicator.

4.3.3 Testing. When testing several instruments in one chamber at the same time, one instrument may be selected to indicate the test rate. This test rate shall be maintained at a constant value while the test readings of all instruments are observed and noted and while the time and altitude interval on the barometer are noted. The computed rate shall be obtained and the scale errors determined.

4.3.4 Calib The standard pressures used in calibrating and testing the indicator shall be as specified in MIL-STD-859.

4.4 Quali cation testing

4.4.1 Test samples. The test samples shall consist of nine indicators representative of the production equipment. Three of the samples shall be subjected to all the tests except reliability and the other six samples shall be subjected to reliability tests only. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. After completion of qualification testing, the contractor shall submit the test samples to the qualifying activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.4.1).

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Quality assurance tests. Quality assurance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each indicator produced on contract or order shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Zero setting

c. Case leakage

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d. Case leakage at -35°C

e. Scale error at room temperature

f. Scale error at -35°C

g. Scale error at +45°C

h. Position error

i. Individual tests of MIL-L-25467 and MIL-L-27160.

4.5.1.1 MIL-STD-105 Sampling and Inspection Procedures shall be applied for Government acceptance of all lots of production indicators on contract. General Inspection Level II of Table I, MIL-STD-105 shall be used throughout the acceptance program. Samples drawn for inspection shall be randomly selected by the Government representative after the whole lot has been submitted for acceptance. The Acceptance Quality Level (AQL) to be maintained is 2.5 using Table IIA of MIL-STD-105 initially. The reduced Sampling Plan of Table IIC may be used when the criteria for reduced sampling has been attained and with approval of the procuring activity. A tightened inspection shall not apply without approval of the procuring activity. Each sample selected shall be subjected to and meet all tests listed under 4.5.1 prior to acceptance of the lot by the Government.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. Three indicators selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.6:

a. Individual tests

b. Pointer lag

c. High temperature operation

d. Low temperature zero error

e. Scale error during temperature change

f. Low temperature scale error operation

g. Zero error during temperature change

h. Vibration error

i. Overpressure

j. Sampling plan A tests of MIL-L-25467 and MIL-L-27160.

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4.5.2.2 Sampling plan B. Unless otherwise specified, (see 6.2), three indicators selected at random from the first 15 and two out of each succeeding 300 indicators or less on the contract or order shall be subjected to the following tests as described under 4.6:

a. Sampling plan A

b. Vibration failure

c. Magnetic effect

d. High altitude-low temperature

e. High temperature exposure

f. Humidity

g. Fungus

h. Salt fog

i. Reliability

j. Sampling plan B tests of MIL-L-25467 and MIL-L-27160.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be acceptable until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.2.3.2 Defective units. A defective unit is an indicator which has failed any test listed under 4.5.1 herein. The unit shall be considered defective as soon as it has failed a test requirement. A unit which has failed any test at only one test point shall be reinspected by the Government at least one time prior to being classified as defective.

4.5.2.3.3 Defective lots. Defective lots will be rejected and returned to the contractor. The lot shall not be resubmitted to the Government for acceptance until, as a minimum, it has been completely reinspected by the contractor. Repairs shall be made as appropriate prior to complete retesting by the contractor. Samples that are subjected to and pass all Government acceptance tests, as specified in 4.5.1, shall not be returned to the contractor with the remainder of the defective lot but shall be accepted as a production item. When resubmitted to the Government for acceptance, the defective lot shall again be subjected to the same acceptance criteria as specified in 4.5.1.1.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 Test methods

4.6.1 Examination of product. Prior to sealing, the indicator shall be inspected to determine compliance with the requirements of 3.3, 3.4.2, 3.7, 3.7.4, 3.10, 3.15, 3.19, 3.20 and 3.21. If these inspections are performed as part of the assembly inspections, they shall be documented during the equipment assembly and this documentation shall be checked as part of the examination of product. After sealing, filling and painting, the indicator shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, markings, weight, dimensions and visibility of the display components. The case shall be inspected for proper painting and general good workmanship and the connector shall be inspected to make sure that no pins are bent or damaged.

4.6.2 Zero setting. The adjusting screw shall be moved throughout its range in both the UP and DOWN directions. The adjustable range of pointer movement shall be from not less than 400 fpm UP and not less than 400 fpm DOWN.

4.6.3 Case leakage. The static pressure connection of the indicator shall be suitably connected to a mercury manometer and sources of vacuum and pressure. An absolute pressure of 4 inches Hg shall be applied to the static pressure connection at a rate not exceeding 20,000 fpm. With the connection tubing pinched off or otherwise completely sealed, the change in the mercury level in the manometer shall not exceed 0.025 inches Hg in 1 minute. Vertical velocity indicators shall be leak tested individually. This procedure shall be repeated with a pressure of 10 inches Hg applied. The same tolerance shall apply.

4.6.3.1 Case leakage at -35°C. The test as specified in 4.6.3 shall be repeated except that the temperature shall be -35°C. The indicator shall be placed in a chamber, cooled to approximately -35°C, and allowed to remain at the -35°C temperature for a period of not less than 3 hours. While at -35°C the case leakage test shall be conducted. The difference in mercury levels shall not exceed 0.05 inches Hg in 1 minute.

4.6.4 Scale error at room temperature. The indicator, with the pointer set to zero, shall be placed in an altitude chamber and a mercury barometer connected to the chamber. The pressure shall be reduced at such a rate that the pointer will indicate an ascent of 500 fpm. While the pointer is indicating 500 fpm, the time required for the mercury barometer to change from a pressure corresponding to an altitude of 1,000 feet to that of an altitude of 1,500 feet shall be determined by a stop watch. The pressure shall then be increased at such a rate that the pointer will indicate a descent of 500 fpm. The time required for the barometer to change from a 1,500-foot altitude to a 1,000-foot altitude shall be determined. This procedure shall be continued for the test rates and altitude as specified in table I. The scale errors shall not exceed the tolerances listed therein.

4.6.4.1 Scale error at -35°C. The indicator shall be subjected to a test specified in 4.6.4 except that during and for 3 hours prior to the test, the temperature shall be -35°C. The scale errors shall not exceed the tolerances as specified in table I.

4.6.4.2 Scale error at +45°C. This test shall be in accordance with 4.6.4.1 except that the temperature shall be +45°C.

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4.6.5 Pointer lag. The static pressure connection of the indicator shall be suitably connected to a water manometer and sources of vacuum and pressure. A vacuum shall be carefully applied to the static port connection. With the indicator showing a rate of ascent, the vacuum applied shall be sufficient to cause the pointer to indicate a rate of descent of approximately 2,500 fpm when the vacuum is instantaneously released by opening the static pressure tubing to the atmosphere and after the pointer has passed through the zero position. The indicator shall be tapped during the return of the pointer to zero. A stop watch shall be started at the instant the pointer passes the 2,000 fpm (descent) graduation and stopped when the pointer passes the 200 fpm (descent) graduation. The time shall be recorded. The procedure shall be repeated, except that a slight pressure shall be carefully applied to the static pressure connection and then released. The time shall be determined for the indication to change from 2,000 fpm climb to 200 fpm climb. The lag of the pointer shall be not less than 4 seconds nor more than 7 seconds. During this test and at any time after the vacuum (or pressure) is released, the pointer shall move smoothly without excessive friction.

4.6.6 High temperature operation. The indicator shall be subjected to a temperature of +71oC for a period of 3 hours. It shall then be allowed to cool to room temperature after which it shall be subjected to the scale error at room temperature and case leakage tests. The indicator shall meet the requirements specified for scale error at room temperature except that the interval and rates as specified in table I shall be used. During and after this test, the setting system shall operate satisfactorily.

4.6.7 Low temperature

4.6.7.1 Low temperature zero error. Before starting this test, the pointer shall be carefully set to zero at room temperature. The indicator shall then be placed in a test chamber, cooled to approximately -54oC within any length of time between 15 minutes and 5 hours, and allowed to remain at the low temperature for a period of not less than 3 hours. With the test chamber at atmospheric pressure, the low temperature zero error during the cooling period and during the 3-hour period at -54oC shall not exceed 150 fpm.

4.6.7.2 Scale error during temperature change. At intervals of 1 hour during the cooling period and during the 3-hour period at -54oC, the indicated rates corresponding to test rates of 2,000 fpm climb and descent, in the standard altitude interval between 2,000 and 4,000 feet, shall be obtained. These indicated rates, corrected for the corresponding zero errors, shall not exceed the values specified in table I.

4.6.7.3 Low temperature operation. After a temperature of -54oC has been maintained within the chamber for 3 hours, the indicator shall be subjected to the test specified for scale error at room temperature except that the intervals and rates as specified in table I shall be used. The indicated rates, corrected for the low temperature zero error, at the end of the 3-hour period at -54oC shall not exceed the values as specified in table I. During and after this test, the setting system shall operate satisfactorily.

4.6.7.4 Zero error during temperature change. After the low temperature scale error test has been conducted, the indicator shall be allowed to return to room temperature at atmospheric pressure by immediate removal from a temperature of -54oC to room temperature. While the temperature of the indicator is increasing, the indication shall be observed for a period of 30 minutes at 1-minute

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intervals for the first 5 minutes and at 5-minute intervals thereafter. During the first 5-minute interval, the maximum reading shall not exceed 300 fpm; thereafter, the maximum reading shall not exceed 200 fpm.

4.6.8 Vibration error. The indicator shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane and with the plane of the mounting lugs of the indicator vertical. The pointer shall be set to zero. No pressure or vacuum shall be applied during this test. The indicator shall be subjected to vibration at an amplitude between 0.003 and 0.005 inch and at frequencies from 500 to 3,000 cpm to determine that the natural frequency of the indicator does not occur in this frequency range. While the indicator is being vibrated, the maximum range of pointer oscillation at any time shall not exceed 50 fpm. Pointer variation from the zero position shall not exceed 50 fpm.

4.6.9 Overpressure. After the pointer has been set to zero, the indicator shall be placed in an altitude test chamber and the pressure decreased at a rate of 60,000 fpm until a pressure corresponding to 50,000 feet is reached. The pressure shall then be increased at a rate of 60,000 fpm until atmospheric pressure is reached. After 5 minutes, the indication of the pointer shall be within 100 fpm of its original indication.

4.6.10 Vibration endurance. With the pointer set at zero, the indicator shall be mounted on a vibration stand as specified for the vibration error test. It shall then be subjected to vibration in a circle having a diameter between 0.018 and 0.020 inch for a 3-hour period. The frequency shall be varied uniformly from 500 to 3,000 cpm and return to 500 cpm once each hour during this test. The difference between the indication of the pointer at the beginning and at the end of the 3-hour vibration period shall not exceed 50 fpm. No looseness in the mechanism or damage to any part of the indicator shall result from this test.

4.6.11 Position error. The pointer shall be set to zero. With atmospheric pressure applied, readings shall be taken while the instrument is being tapped lightly in each of several different positions. The change in indication shall not exceed 50 fpm.

4.6.12 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from and magnetically East or West of the center of a short bar magnet compass. The horizontal magnetic field intensity shall be 0.17° to 0.19° oersted. The deflection of the compass shall not exceed 1°.

4.6.13 High altitude - low temperature. The indicator shall be placed in an altitude chamber maintained at a temperature of -65°C and the pressure reduced to simulate an altitude condition of 50,000 feet (3.436 inches Hg absolute pressure) for a period of 4 hours. The indicator shall then be maintained at a temperature of -65°C, atmospheric pressure, for an additional period of 44 hours. After returning to room temperature the indicator shall be carefully examined for any evidence of damage. No damage to any part of the indicator shall result from this test.

4.6.14 High temperature exposure. The indicator shall be subjected to a temperature of +71° ±2°C for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the indicator shall operate without indication of sticking or erratic action. After the chamber has returned to room temperature and the indicator has been maintained at room

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temperature for a minimum period of 4 hours, it shall meet the individual tests. There shall be no damage that would adversely affect subsequent operation.

4.6.15 Humidity. The indicator shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.1, procedure I. To simulate installed conditions, the external connections to the indicator shall be connected with 10 feet of .025-inch copper or aluminum tubing so coiled that the moisture can drain out of the open end. Within 1 hour after the 240-hour period, the indicator shall meet the scale error at room temperature test.

4.6.16 Fungus. The indicator shall be subjected to a fungus resistance test in accordance with MIL-STD-810, Method 508.1, procedure I, except that the test period shall be 14 days. The installed conditions of the humidity test shall apply.

4.6.17 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1, procedure I. The installed conditions of the humidity test shall apply. At the end of the 50-hour test period, the indicator shall be subjected to and shall meet the individual tests.

4.6.18 Reliability. Each indicator shall be subjected to a minimum pressure cycling of 3 cycles per hour for a minimum of 880 hours, or 2,640 cycles consisting of a pressure variation equivalent to at least three-fourths of full scale. Each pressure cycle shall consist of operation to the maximum required pressure and return to ambient. The maximum scale value shall be maintained for not less than 2 minutes. Approximately 2 minutes at ambient pressure shall be allowed between each pressure cycle. During the fifth cycling period, the indicators shall be vibrated at an amplitude of 0.009 to 0.011 inch and a frequency of 35 Hz. The cycling period, number of hours or cycles, and cycling temperature shall be accomplished as specified in table II. After each period of cycling, the indicator shall be subjected to and shall meet the room temperature scale error test, each out of tolerance condition shall be considered as a malfunction. Upon completion of the final cycle, the indicator shall then be subjected to the individual tests. Failure to meet any one of the individual tests shall be considered a malfunction. In case of malfunction of any kind, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. If recalibration is required to pass the scale error test, an additional malfunction shall be charged against this test. This test shall demonstrate a minimum acceptable MTBF of 1,200 hours to a 90 percent confidence level. The level of reliability shall be verified when the six indicators have satisfactorily complied with the requirements of table III.

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TABLE II. Cycling.

Cycling Period	Hours or Cycles		Ambient Temp °C
1	150	450	25
2	100	300	55
3	150	450	25
4	100	300	-35
5	100	300	25
6	100	300	55
7	100	300	-35
8	100	300	25

TABLE III. Reliability verification.

M (Maximum Number of Malfunctions Corresponding to the Specified T)	(Total Test Time) Hours or Cycles	
0	5,280	15,840
1	6,240	18,720
2	7,200	21,600
3	8,160	24,480
4	9,120	27,360
5	10,200	30,600

4.6.20 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471 for class ID equipment.

4.6.21 Lighting. Lighting tests shall be conducted in accordance with MIL-L-25467 and MIL-L-27160.

4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

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4.7.2 First article. When the unit container is capable of serving as the shipping container first article inspection, and rough handling tests as outlined in section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A, B, or C as specified by procuring activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density, and thickness that will protect the unit from impact shock damage when tested as specified in 4.7.2 of this specification.

5.1.2 Level B. Each unit shall be packaged as indicated for Level A except that a PPP-B-636 class domestic box shall be used as the unit container.

5.1.3 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with appendix of PPP-B-636.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The AAU-18/A indicator covered by this specification is intended for use on aircraft to indicate the vertical component of the aircraft speed based on the rates of change of atmospheric pressure.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. Levels of packaging and packing required

d. Data.

6.3 Definitions

6.3.1 Tapping. The phrase TAPPED LIGHTLY means the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 Amplitude. Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

6.3.3 Test reading. A test reading is the indication of the instrument during the test.

6.3.4 Test rate. The test rate means the test point settings, specified in column 1 of table I, at which the instrument is to be tested.

6.3.5 Computed rate. The computed rate is the rate computed by dividing the altitude interval (2,000 feet as measured by a manometer) by the time in seconds (required for the change in altitude) and multiplying by 60. For example:

$$\text{Computed rate} = \frac{\text{Altitude interval} \times 60}{\text{Time in seconds}}$$

6.3.6 Computed rate correction. The computed rate correction is the algebraic difference between the test rate and computed rate.

Example 1:	Test rate	+2000
	Computed rate	+1925
	Computed rate correction	+75
Example 2:	Test rate	+2000
	Computed rate	+2025
	Computed rate correction	-25

6.3.7 Indicated rate. The indicated rate is the reading of the instrument as computed herein to correspond to a test rate. The indicated rate of the instrument should be computed by adding algebraically the test reading of the indicator and the computed rate correction.

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6.3.8 Scale error. Scale error is the difference between the test rate and the indicated rate.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order thay they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASD/ENAID, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4.1 Test report. The contractor will submit three copies of the test report specified in 4.4.2 to the procuring activity for approval.

6.5 International standardization. Certain provisions (3.4.1 and 3.7.4) of this specification are the subject of international standardization agreement STANAG 3320. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Officers so that appropriate action may be taken respecting the international agreement concerned.

6.6 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 99

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International Interest (See 6.5)

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